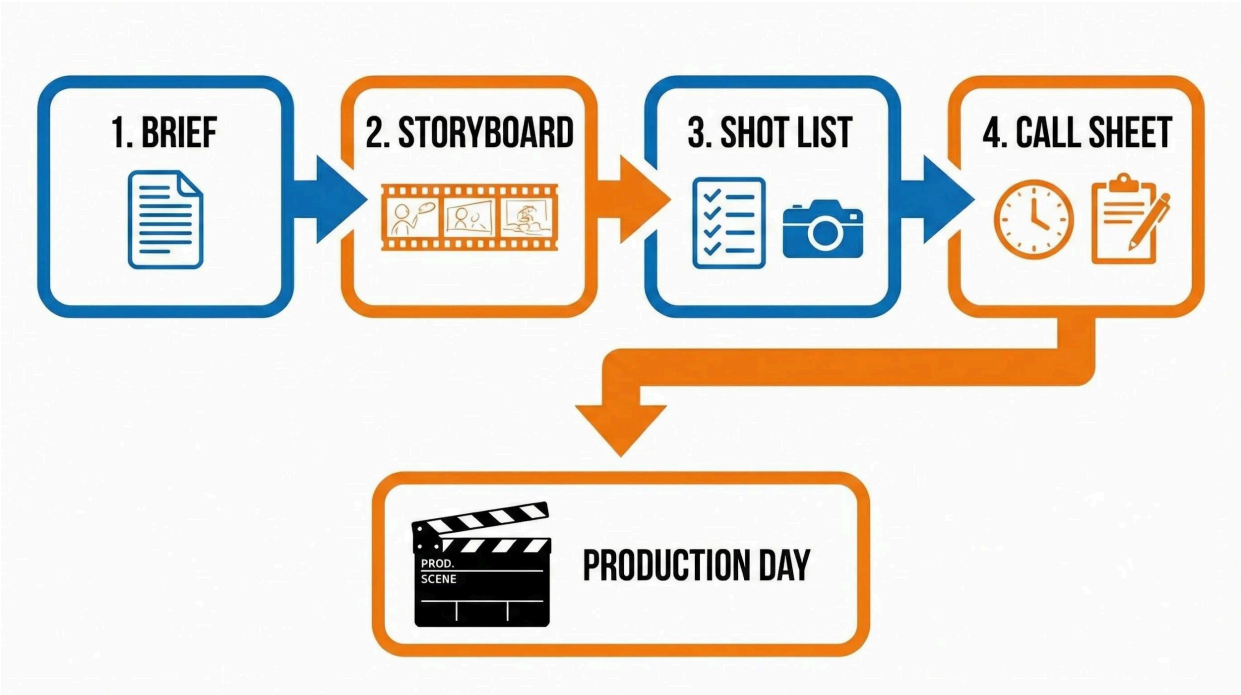
70MM AI — Four-Group Major Project Architecture

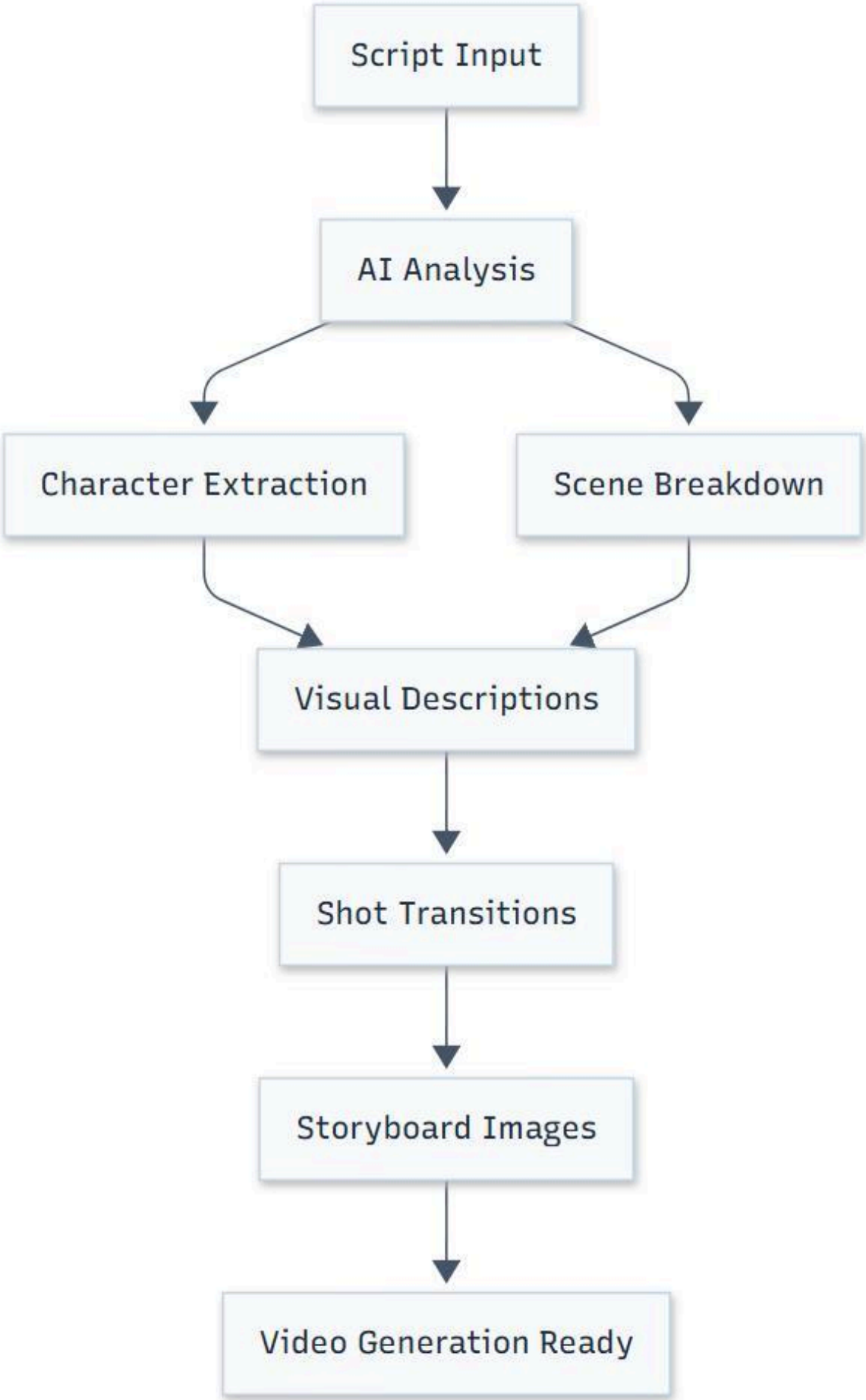



1. BRIEF
2. STORYBOARD
3. SHOT LIST
4. CALL SHEET
PROD.
SCENE
PRODUCTION DAY

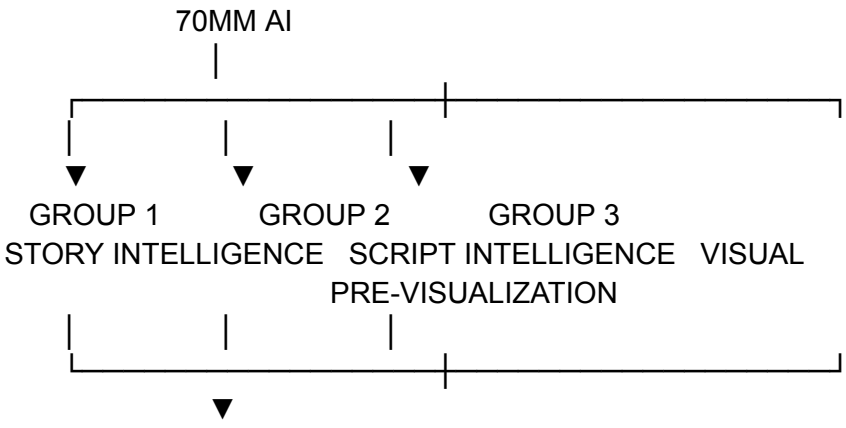

SCRIPT TO STORYBOARD
VISUALIZE • PLAN • TELL YOUR STORY
THUMBNAIL IDEAS
1 THE SCRIPT
THE FOUNDATION OF THE STORY
2 THUMBNAILING
PLANNING SHOTS, COMPOSITION & FLOW
3 FINAL STORYBOARD
THE VISUAL BLUEPRINT FOR PRODUCTION


SHOTLIST
1. WIDE ESTABLISHING SHOT
2. TWO SHOT - DIALOGUE
3. OVER THE SHOULDER
4. TRACKING SHOT - FOLLOW
5. CLOSE UP - REACTION
6. ACTION BEAT
7. WIDE - AFTERMATH
EMOTIONAL ARC
- TENSION
- CONFRONTATION
- RESOLUTION
LOCATIONS
- WAREHOUSE
- ALLEY
- ROOFTOP


Size | Subject(s) | MVT/support | Notes
WS | J & L | Dolly - in/out | whole scene
CU | J | Ped down (gimbal) | face
MCU | J & L | rack focus | J → L
Med. | J | handheld | clean single
Med. | L | sticks | box in FG
Med | L | sticks | clean si
MWS | J & L | sticks | med. 2-
CU | L | sticks | profile
XCU | L | sticks | eyes
MS | J | tracking/gimbal | J gets off couch
MWS | J | sticks | profile cam right
CU | box | sticks | J POV (hands)
Med. | J | dolly R→L | OTS L
CU | L | dolly zoom | L w/ popcorn
CU | J | sticks HH | box in FG
Med. | popcorn | on floor | slow-motion
wide 2-shot
3/4 J
3/4 L
single J
single L
low-angle pkg. FG


Script Input
AI Analysis
Character Extraction
Scene Breakdown
Visual Descriptions
Shot Transitions
Storyboard Images
Video Generation Ready


SCENE 73 - TAKE 1
73


70MM AI
GROUP 1
STORY INTELLIGENCE
GROUP 2
SCRIPT INTELLIGENCE
GROUP 3
VISUAL
PRE-VISUALIZATION

GROUP 4
AI PRE-PRODUCTION
& PRODUCTION PLANNING

The four groups should **share a common AI platform**, but each group should own a clearly defined research/engineering problem.

GROUP 1 — AI STORY INTELLIGENCE ENGINE(Somanadry and team)

Core problem

> **Given a user's idea, story, or screenplay concept, can AI help develop, analyze, structure, evaluate and improve the story according to cinematic storytelling principles?**

This should be your **Story Development + Story Analysis + Film Knowledge Intelligence** module.

---

1. Idea Generation Engine

The user can start with something as simple as:

> "A college student discovers that his dead father was secretly a spy."

The system should help expand this into:

Idea → Premise → Logline → Conflict → Characters → Plot → Sequences → Events → Story

Features

- Idea generation
- Premise generation
- Logline generation
- Theme generation
- Central conflict generation
- Story-world generation

- What-if generation
- Alternative plot generation
- Multiple endings
- Twist generation
- Subplot generation
- Narrative hook generation

Advanced feature

Idea Expansion / Compression

One-line idea
↓
Premise
↓
Logline
↓
Short synopsis
↓
Detailed synopsis
↓
Full story

---

2. Plot Development Engine

The AI should understand cinematic narrative structures.

It can generate:

- Plot points
- Story beats
- Sequences
- Events
- Turning points
- Inciting incident
- Midpoint
- Crisis
- Climax
- Resolution

Support multiple structures rather than forcing everything into Hollywood's three-act structure.

Possible structures

- Three-act structure
- Five-act structure
- Hero's Journey
- Save the Cat
- Kishōtenketsu
- Non-linear narrative
- Circular narrative
- Parallel narrative
- Anthology
- Episodic structure
- Investigation structure
- Thriller structure

This is where **cinematic narrative modelling** becomes important.

---

3. Story Evaluation Engine

This is one of the most valuable research components of Group 1.

If the user already has a story, the system should **not simply rewrite it**.

It should diagnose it.

Story Doctor / Story Analyst

Evaluate:

- Premise strength
- Originality
- Narrative coherence
- Character motivation
- Conflict
- Stakes
- Pacing
- Emotional progression
- Theme
- Act structure
- Scene progression
- Payoff
- Setup

- Foreshadowing
- Resolution
- Genre consistency

Then produce:

> **Score + diagnosis + recommendations**

Example:

Narrative Structure 7.8/10
Character Development 6.2/10
Conflict 8.1/10
Pacing 5.9/10
Emotional Arc 7.3/10
Originality 8.0/10
Genre Consistency 6.8/10
Ending 5.7/10

---

4. Genre Intelligence

The system should understand that a **thriller is not evaluated in exactly the same way as a romantic comedy**.

Possible genres:

- Drama
- Romance
- Comedy
- Romantic comedy
- Action
- Thriller
- Crime
- Mystery
- Horror
- Psychological
- Sci-fi
- Fantasy
- Historical
- Political
- Social drama
- Coming-of-age
- Experimental

The system should identify:

> **Primary Genre + Secondary Genre + Genre Expectations**

Then evaluate whether the story satisfies those expectations.

---

5. Film Reference Intelligence

This is where your idea becomes much more interesting.

The system can use a **licensed/curated film knowledge base** rather than indiscriminately copying films.

The user can ask:

> "What films have similar narrative structures?"

The system retrieves structurally similar references.

For example:

User Story
↓
Semantic Embedding
↓
Film Knowledge Base
↓
Similarity Retrieval
↓
Similar Films / Structures / Tropes

Use:

- NLP embeddings
- Vector database
- RAG
- LLM reasoning

The output should focus on:

structure, themes, tropes, pacing and narrative devices, not copying dialogue or copyrighted screenplay text.

---

6. Story Improvement Engine

After analysis:

> "What should I change?"

The AI can suggest:

- Stronger inciting incident
- Character motivation
- Better midpoint
- Stronger antagonist
- Additional conflict
- Subplot
- Better climax
- Foreshadowing
- Emotional payoff
- Pacing changes

Group 1 final pipeline

IDEA
↓
IDEA GENERATION
↓
PREMISE
↓
LOGLINE
↓
PLOT
↓
SEQUENCES
↓
EVENTS
↓
STORY
↓
STORY ANALYSIS
↓
GENRE ANALYSIS
↓
FILM REFERENCE ANALYSIS

↓
STORY IMPROVEMENT

GROUP 2 — AI SCRIPT & CHARACTER INTELLIGENCE ENGINE(Chaitanya and team)

Group 2 starts **after the story is ready**.

Its central question:

> **How do we transform a story into a professionally structured screenplay and help the writer develop it?**

---

1. Story → Screenplay Conversion

Input:

> Full story

Output:

> Screenplay

The model should identify:

- Scenes
- Sluglines
- Action
- Characters
- Dialogue
- Parentheticals
- Transitions

Example:

STORY
↓

SCENE EXTRACTION
↓
SCENE ORDERING
↓
CHARACTER EXTRACTION
↓
DIALOGUE GENERATION
↓
SCREENPLAY FORMAT

---

2. Professional Screenplay Editor

This should be a **major feature**, not an ordinary text editor.

Think:

> **Celtx + Arc Studio + AI**

The editor should support:

- Scene headings
- Action
- Character
- Dialogue
- Parenthetical
- Transition
- Shot descriptions where appropriate
- Page formatting
- Scene numbering
- Revision marks
- Character tracking

AI-assisted writing

While writing:

> "Continue this scene."
>
> "Make this dialogue more natural."
>
> "Give this character more subtext."
>
> "Convert this scene to Telugu-English Tenglish."

---

3. Tenglish / Hinglish NLP

This should become one of 70MM AI's distinguishing features.

Support:

English

> "Where were you last night?"

Tenglish

> "Nuvvu last night ekkada unnava?"

Hinglish

> "Tum kal raat kahaan the?"

The system should understand **code-switching**, not merely translation.

It should preserve:

- Cultural context
- Character personality
- Natural speech
- Regional expressions
- Slang
- Emotional tone

Technical terminology

This is essentially:

> **Multilingual / Code-Switched NLP**

with:

> English + Telugu + Tenglish + Hindi + Hinglish

as your primary language scope.

---

4. Character Intelligence

The system should automatically construct a:

Character Bible

For every character:

- Name
- Age
- Gender
- Occupation
- Personality
- Background
- Motivation
- Goal
- Fear
- Strength
- Weakness
- Relationships
- Character arc
- Emotional state
- Dialogue style
- Speech patterns

---

5. Character Development Engine

The writer can ask:

> "This character feels weak. How can I improve him?"

The system should analyse:

- Character motivation
- Internal conflict
- External conflict
- Arc
- Agency
- Relationships
- Transformation

Then suggest alternatives.

---

6. Script Doctor

This should be a dedicated AI subsystem.

Script Doctor analyses:

- Weak scenes
- Redundant scenes
- Poor dialogue
- Exposition
- Pacing
- Character inconsistency
- Plot holes
- Missing setup/payoff
- Weak transitions
- Repetitive dialogue
- Unrealistic character behaviour
- Genre violations

Then:

> **Diagnosis → Recommendation → Optional Rewrite**

---

7. Script Coverage

Generate professional-style:

- Logline
- Synopsis
- Character breakdown
- Genre
- Strengths
- Weaknesses
- Market positioning
- Structural analysis
- Writer recommendations

- Overall score

Group 2 pipeline

STORY
↓
STORY → SCRIPT
↓
SCREENPLAY EDITOR
↓
CHARACTER BIBLE
↓
CHARACTER DEVELOPMENT
↓
SCRIPT ANALYSIS
↓
SCRIPT DOCTOR
↓
SCRIPT COVERAGE
↓
FINAL SCREENPLAY

GROUP 3 — AI STORYBOARD & CINEMATOGRAPHY ENGINE (Komal and team)

This should probably be the **most visually impressive module** of the entire 70MM project.

Central problem:

> **How do we convert a 120–130 page screenplay into a coherent visual shooting plan and storyboard?**

And you correctly identified the difficult part:

120–130 pages ≈ feature-length screenplay

The system must not treat every page independently.

It needs **hierarchical screenplay understanding**.

---

1. Screenplay Parsing

Input:

> 120–130 page screenplay

The model should extract:

SCREENPLAY
↓
ACTS
↓
SEQUENCES
↓
SCENES
↓
BEATS
↓
SHOTS

This hierarchy is critical.

---

2. Scene Understanding

For each scene identify:

- Characters
- Location
- Time
- Action
- Emotion
- Conflict
- Important objects
- Dialogue
- Visual emphasis
- Continuity requirements

---

3. Automatic Shot Division

This is a major research problem.

The AI proposes:

- Establishing shot
- Wide shot
- Medium shot
- Medium close-up
- Close-up
- Extreme close-up
- Two-shot
- Over-the-shoulder
- POV
- Insert
- Cutaway

and determines **why each shot exists**.

---

4. AI Storyboard Generation

This is the **flagship feature**.

SCREENPLAY
↓
SCENE
↓
SHOT PLAN
↓
VISUAL DESCRIPTION
↓
IMAGE GENERATION MODEL
↓
STORYBOARD

The important challenge isn't simply generating beautiful images.

It is:

> **Maintaining visual and character consistency across hundreds of generated storyboard frames.**

---

5. Character Consistency

If Rahul appears in Scene 4:

> Rahul = same visual identity

Then in Scene 70:

> Rahul should still look like Rahul.

This requires:

- Character reference images
- Identity embeddings
- Image-conditioning
- Consistent generation
- Character asset management

---

6. Cinematography Intelligence

For every shot:

Camera

- Shot size
- Camera angle
- Lens
- Camera height
- Camera movement
- Framing
- Composition

Lighting

- Key light

- Fill
- Backlight
- Practical lights
- Hard/soft light
- Direction
- Intensity
- Color temperature

Visual language

- Mood
- Color palette
- Contrast
- Depth of field
- Perspective

---

7. AI Storyboard Editor

Users should be able to modify:

> "Make this a low-angle shot."
>
> "Change to night."
>
> "Move camera behind the character."
>
> "Use a 50mm lens."
>
> "Make the lighting dramatic."

The image-generation model regenerates the frame.

---

8. Visual Continuity

The system should compare storyboard frames and detect:

- Character inconsistency
- Costume inconsistency
- Prop inconsistency

- Location inconsistency
- Lighting inconsistency
- Screen direction errors
- Spatial continuity issues

---

9. Storyboard → Shot List

Automatically produce:

Scene	Shot	Type	Lens	Movement	Lighting	Characters

This connects Group 3 directly with Group 4.

Group 3 pipeline

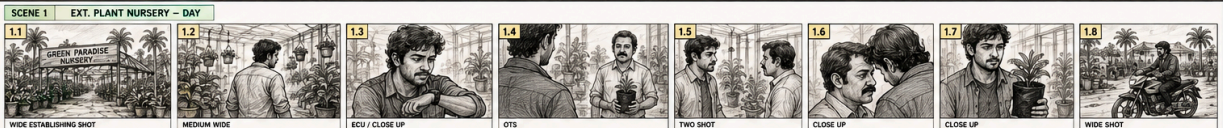
120–130 PAGE SCREENPLAY
↓
SCREENPLAY PARSER
↓
ACTS
↓
SEQUENCES
↓
SCENES
↓
BEATS
↓
SHOT DIVISION
↓
CINEMATOGRAPHY PLAN
↓
IMAGE GENERATION
↓
STORYBOARD
↓
VISUAL CONTINUITY
↓
SHOT LIST

BOARD 1 : SCENES 1 – 5
NURSERY TO ROOM ARRIVAL

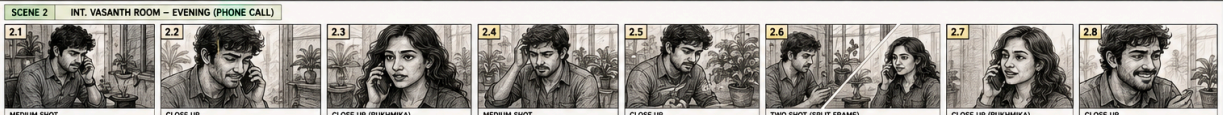
RUKHMIKA VASANTH LA 50TH BREAK UP

LENGTH : ~ 22 – 24 MINS (PORTION) | PAGES : 1 OF 4

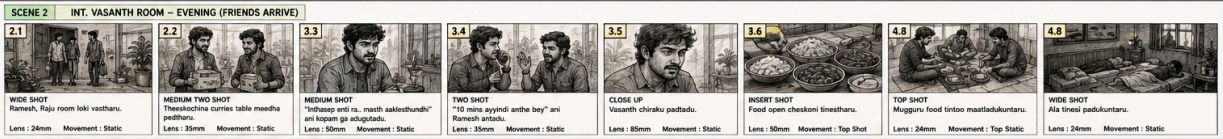
SCENE 1 EXT. PLANT NURSERY – DAY



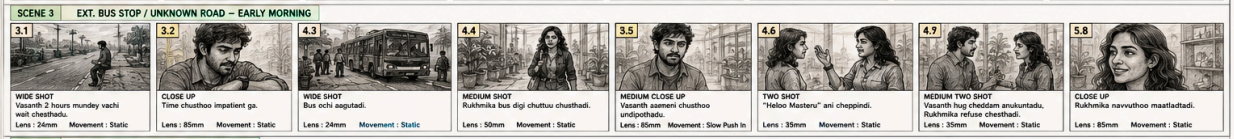
SCENE 2 INT. VASANTH ROOM – EVENING (PHONE CALL)



SCENE 2 INT. VASANTH ROOM – EVENING (FRIENDS ARRIVE)



SCENE 3 EXT. BUS STOP / UNKNOWN ROAD – EARLY MORNING



SCENE 5 INT. VASANTH ROOM – DAY (ARRIVAL)



CAMERA STYLE : Natural, grounded | COLOR TONE : Warm daylight to evening natural | LIGHTING : Natural light / practical lamps | MOOD : Warm, Nostalgic, Excited | NOTES : Establishing characters, tone and world. | LENS GUIDE : 24mm (Wide) 35mm (Normal) 50mm (Natural) 85mm (Close Up)

BOARD 4 SCENES 17 – 21 :
FINAL BREAKUP TO PATCHUP & ENDING

RUKHMIKA VASANTH LA 50TH BREAK UP

LENGTH : ~ 24 – 26 MINS (PORTION) | PAGES : 4 OF 4

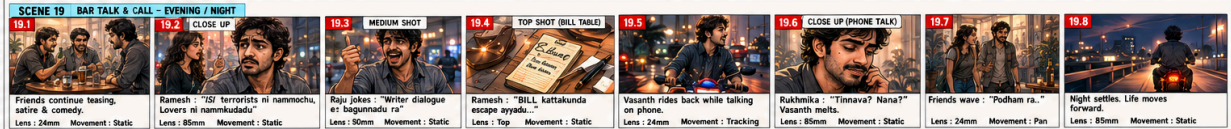
SCENE 17 FINAL BREAKUP – AFTERNOON



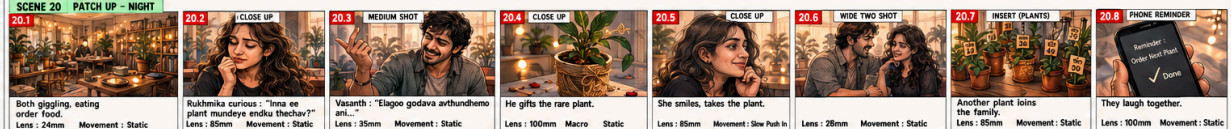
SCENE 18 PLANT AGREEMENT REVEAL – EVENING



SCENE 19 BAR TALK & CALL – EVENING / NIGHT



SCENE 20 PATCH UP – NIGHT



SCENE 21 ENDING MONTAGE – NIGHT (FINAL)



CAMERA STYLE GUIDE : Naturalistic, emotional, intimate, warm.

COLOR TONE : Warm, cozy, nostalgic, with practical lights.

LIGHTING : Window light, table lamps, fairy lights, natural night.

MOOD : Romantic, playful, emotional & hopeful.

LENS GUIDE : 24mm (Wide) 35mm (Normal) 50mm (Natural) 85mm (Close Up) 100mm (Macro)

NOTES
- Plant = Truth, Love, Memory
- Breakup = Ego, Insecurity, Truth
- Patchup = Realization, Growth, Love

GROUP 4 — AI PRE-PRODUCTION & PRODUCTION PLANNING ENGINE(Sireesha and team)

Group 4 begins with:

> **Final screenplay + shot list**

and answers:

> **"How do we actually shoot this film?"**

---

1. Automated Script Breakdown

The system reads the entire screenplay and generates a:

> **Production Breakdown**

Extract:

Cast

- Lead characters
- Supporting characters
- Minor characters
- Extras
- Crowd requirements

Locations

- Houses
- Offices
- Roads
- Hotels
- Colleges
- Hospitals
- Studios
- Outdoor locations

Props

- Vehicles
- Phones
- Weapons where applicable
- Furniture
- Documents
- Food
- Special props

Costume

- Character
- Scene
- Costume
- Continuity

Makeup / Hair

- Requirements
- Special makeup
- Age makeup
- Continuity

---

2. Equipment Breakdown

Automatically estimate:

Camera

- Camera body
- Lenses
- Tripod
- Gimbal
- Handheld setup
- Drone where applicable

Lighting

- Key lights
- Fill lights

- Backlights
- LED panels
- Practical lighting

Grip

- Tripods
- Stands
- Sliders
- Dollies
- Rigging

---

3. Location Breakdown

For every scene:

Scene 24

Location: Railway Station
Time: Night
Characters: 4
Extras: 20
Props: 8
Vehicles: 2
Lighting: Night exterior
Equipment: Dolly + 50mm + 85mm

---

4. Budget Estimation Engine

This becomes an important applied-ML component.

Calculate:

Pre-production

- Location
- Casting
- Costume
- Props

Production

- Camera rental
- Lens rental
- Lighting
- Equipment
- Crew
- Transport
- Food
- Accommodation
- Location charges

Post-production

- Editing
- Sound
- Music
- VFX
- Color grading

Then:

> **Estimated Production Budget**
>
> **Ref**-
> https://docs.google.com/document/d/1Jrb0zhHXo-5n8xOZU6j-tL14EhvmHLMXJIIoVtI7_vM/edit?usp=sharing/

---

5. Production Scheduling

This is another major feature.

Input:

- Script
- Available actors
- Locations
- Equipment
- Shooting dates
- Budget constraints

Output:

Optimized Shooting Schedule

For example:

DAY 1
Location: College

Scenes: 4, 8, 12, 17, 23

Actors:
A
B
C

Equipment:
Camera A
24mm
50mm
Gimbal

Call Time: 06:00
Wrap: 18:00

---

6. Location-Based Scheduling

Instead of shooting chronologically:

Scene 1
Scene 2
Scene 3
Scene 4

the AI should optimize:

Shoot all scenes belonging to the same location together.

This can significantly reduce:

- Location cost
- Transport
- Setup time

- Equipment movement

---

7. Actor Availability Optimization

If:

Actor A available: Days 1–5
Actor B available: Days 3–8
Actor C available: Days 2,4,7

the system should generate a feasible schedule.

This becomes an **optimization problem**, rather than simply an LLM task.

---

8. Production Calendar

Generate:

- Shooting days
- Scene allocation
- Location allocation
- Actor allocation
- Equipment allocation
- Call sheets
- Daily production plan

---

9. Call Sheet Generator

For each shooting day:

- Date
- Location
- Call time
- Cast
- Scenes

- Equipment
- Costume
- Makeup
- Transport
- Production notes
- Emergency/contact information where appropriate

---

10. Production Risk Analysis

This is one feature I would add.

AI can identify:

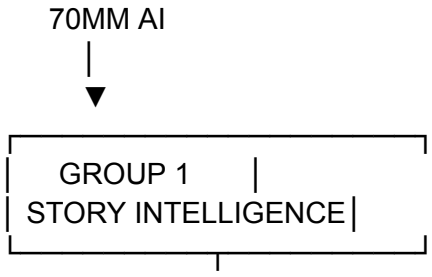
- Too many locations
- Excessive night shoots
- Large cast requirements
- Difficult weather-dependent scenes
- Expensive sequences
- Complex equipment requirements
- Crowd scenes
- Difficult stunts/VFX
- Schedule conflicts

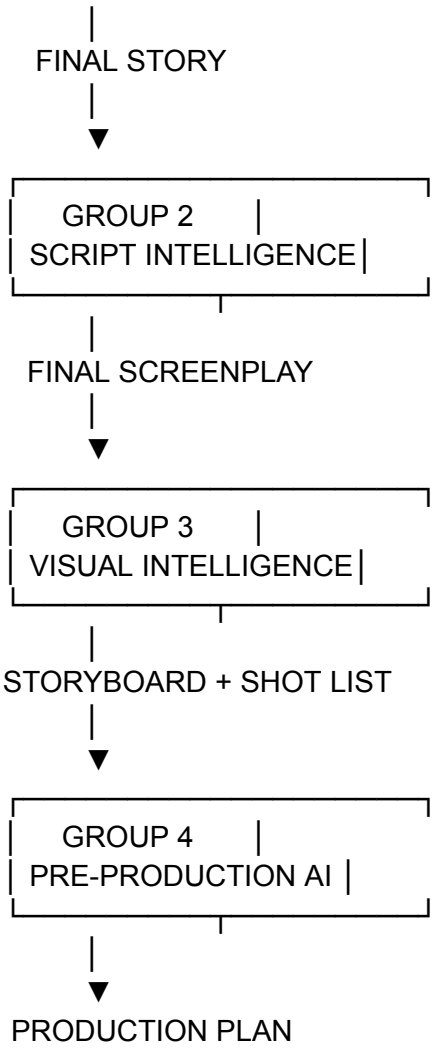
Then:

Production Risk Score

Complete 70MM AI Architecture

Now the four groups become extremely clean:





AI Technology Stack

I would **not** make every component an LLM. That would weaken the project technically.

Use the appropriate model for each problem.

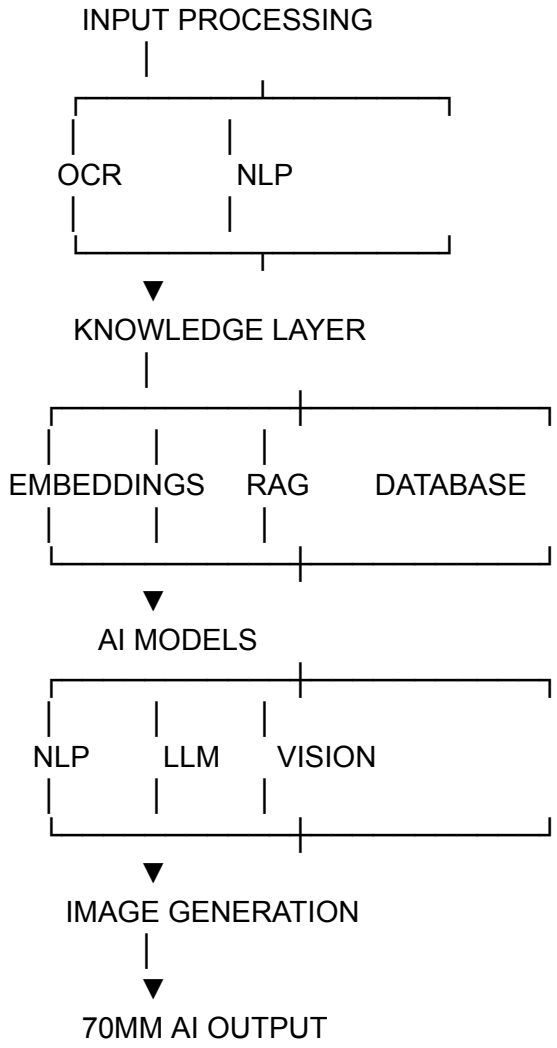
Requirement	Appropriate AI
Story generation	LLM + NLP
Story analysis	LLM + NLP + embeddings
Genre classification	Transformer / fine-tuned NLP
Film similarity	Embeddings + vector DB

Story structure	NLP + LLM
Tenglish/Hinglish	Multilingual Transformer + LLM
Screenplay parsing	NLP + LLM
Character extraction	NER + LLM
Script formatting	NLP + deterministic parser
Character consistency	Embeddings + LLM
Storyboard generation	Diffusion/Image Generation
Character consistency	Image conditioning + embeddings
Shot classification	Computer Vision
Scene understanding	Vision-Language Model
Cinematography recommendation	Multimodal model + rule/knowledge layer
OCR	Deep-learning OCR
Script PDF extraction	OCR + NLP
Production breakdown	NLP + LLM
Budget estimation	ML + rule-based calculations
Scheduling	Optimization algorithms + ML
Actor availability	Constraint optimization
Location scheduling	Constraint optimization
Production risk	ML + rules + LLM

Most Important: Don't Call It "Only LLM"

For the academic project, your architecture should look more like:





Model categories you should investigate

NLP / LLM

Use models suitable for:

- English
- Telugu
- Hindi
- Code-switching
- Long-context screenplay understanding

The **Tenglish/Hinglish capability should be a specific research component**, not merely a translation button.

Vision

Use:

- Vision Transformers
- CNNs where appropriate
- Vision-Language Models
- Object detection
- Face/character recognition
- Scene recognition
- Image embeddings

Image Generation

Use diffusion/modern image-generation models for:

- Storyboards
- Character visualization
- Location visualization
- Shot visualization
- Lighting visualization

OCR

For:

- Uploaded screenplay PDFs
- Scanned scripts
- Handwritten notes where feasible
- Production documents

One Critical Technical Challenge: 130-Page Screenplay

I would make this an explicit research problem for **Group 3**, rather than simply giving the students an LLM API and asking it to "generate storyboard."

A 120–130 page screenplay cannot safely be treated as:

> **one prompt → one output**

Instead:

130 PAGE SCRIPT
↓
Document OCR / Parsing
↓
Chapter / Act Detection
↓
Sequence Detection
↓
Scene Detection
↓
Beat Detection
↓
Character / Location Tracking
↓
Scene Graph
↓
Shot Planning
↓
Storyboard Generation
↓
Continuity Verification
↓
Complete Storyboard

The **Scene Graph / Narrative Graph** should become a shared representation between Groups 1, 2 and 3.

That is what will allow 70MM AI to maintain consistency across a feature-length film.

Division of Deliverables

Group 1 — Story Intelligence

Primary deliverable: AI Story Development & Story Doctor

> Idea → Plot → Sequences → Events → Story → Analysis → Genre Evaluation → Film References → Improvement

Group 2 — Script Intelligence

Primary deliverable: AI Screenwriting Studio

Story → Screenplay → Script Editor → Character Bible → Character Development → Script Doctor → Script Coverage

Group 3 — Visual Intelligence

Primary deliverable: AI Storyboard & Cinematography Studio

Screenplay → Scene Understanding → Shot Division → Cinematography → Lighting → Image Generation → Storyboard → Shot List

Group 4 — Production Intelligence

Primary deliverable: AI Pre-Production & Production Planner

Screenplay → Cast → Characters → Locations → Props → Costumes → Equipment → Budget → Scheduling → Call Sheets → Production Plan

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The four groups should **not develop four separate applications**.

They should develop **four AI engines connected through a common 70MM AI data model**:

